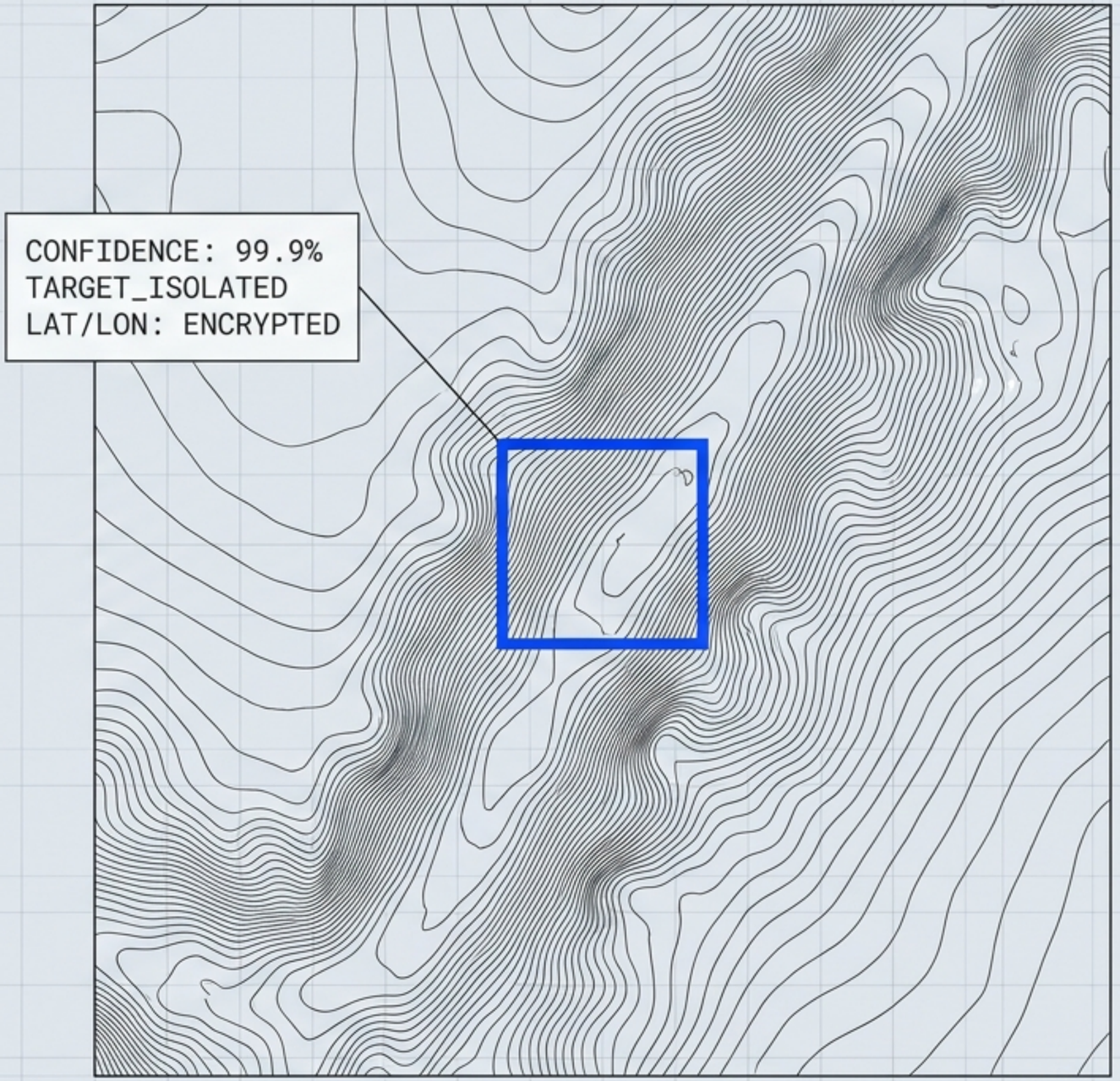
PHYS-UNMASK: The Blue Ocean SAR Engine

Reprocessing historical satellite data with deterministic physics for 2030-era forensic recovery.



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THE ELEGANT HACK: WE DON'T NEED NEW FLIGHTS

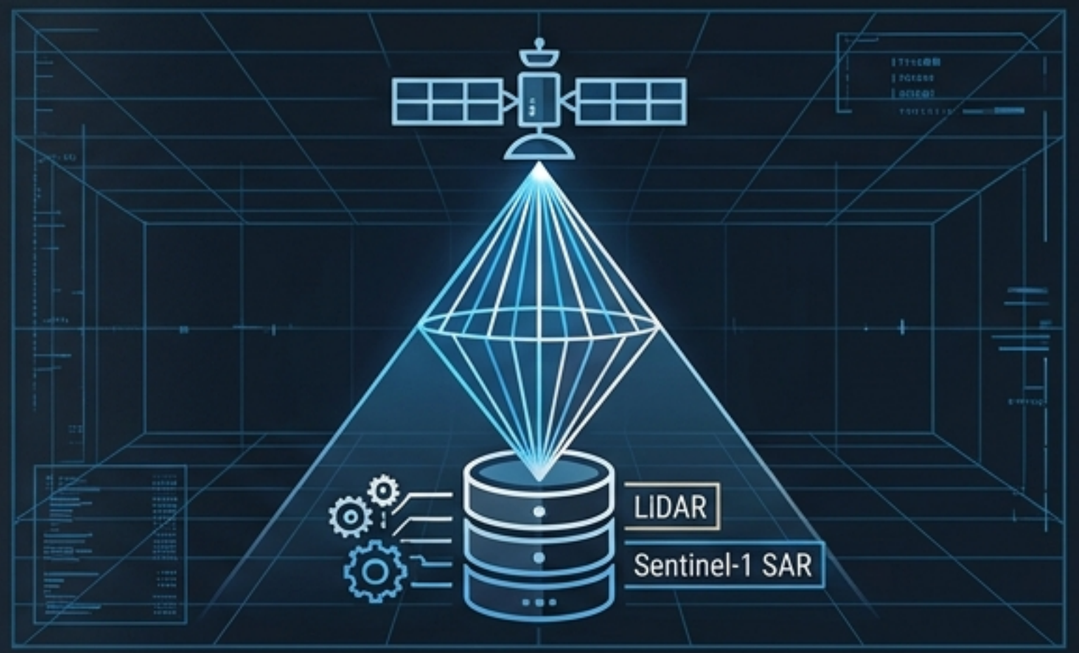
Wilderness search and rescue (WiSAR) has hit physiological limits. Incident volumes are surging, and human volunteer persistence is no longer scalable. The reflexive solution is to fly new drone grids—but the environment inherently fights back.

THE ILLUSION OF NEW



New flight acquisitions are blinded by chaotic weather, canopy occlusion, and time delays. We are fighting the environment.

THE DETERMINISTIC REALITY



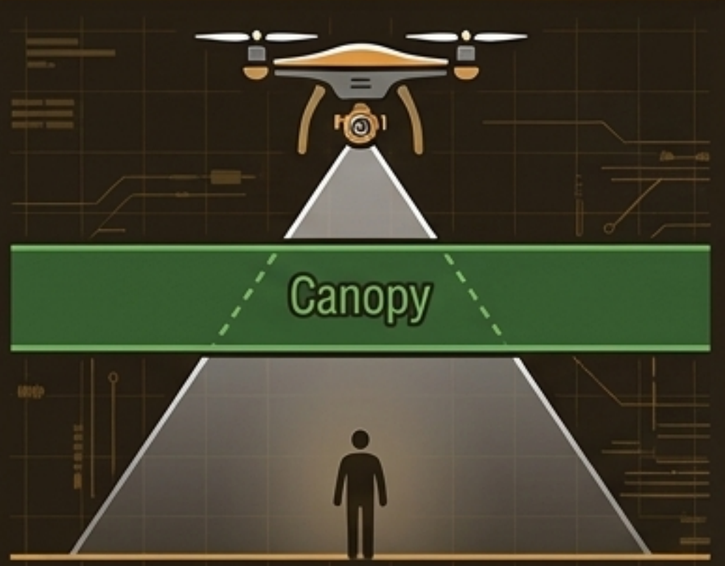
The satellites already saw it. By applying brute-force, deterministic physics mathematics to existing historical multi-temporal LiDAR and Sentinel-1 SAR data, we can retroactively unmask the terrain. The data exists; we just lacked the math.

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THE FAILURE OF THE NAKED EYE & LEGACY CODE

Wilderness search and rescue (WISAR) has hit physiological limits. Incident volumes and building ɪrrent up persistence, and human volunteers are a oʊsistant. The recine solution to ſnaurne grids—compodity dismoever the, effortness solutions.

⚠ OPTICAL BLINDNESS ⚠



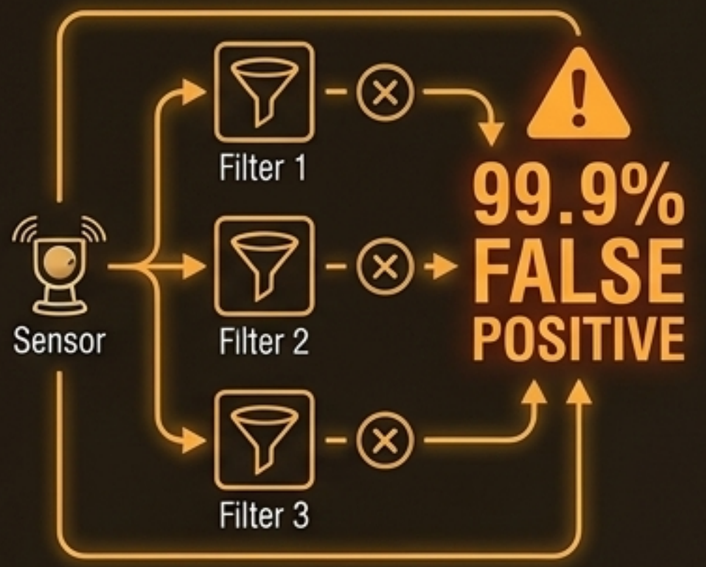
4K cameras cannot penetrate dense evergreen foliage; they merely map the treetops. A drone cannot see into a shadowed ravine at the angles required to identify a human shape.

THE EBBY STEPPACH EFFECT



Buried remains in utility vaults vent decomposition gases horizontally through underground pipe networks (up to 193m downstream). This prevents vertical gas percolation, causing 0% surface K-9 scent alerts directly overhead.

NAÏVE BAYES DOUBLE-COUNTING



Legacy heuristic algorithms extract multiple feature passes from a single sensor and multiply them as independent evidence. This double-counting artificially inflates confidence and triggers severe false positives.

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GROUND TRUTH EXTRACTION: STRIPPING THE ROOTBALL

Wilderness search and rescue (WISAR) has hit physiological limits. Incident volumes and building ɪrent up persistence, and human volunteers are a ovsistant. The recine solution to fnaurne grids—compodity dismoever the, effortness solutions.

LEGACY SMRF FILTER



Legacy algorithms misclassify dense elevated root structures as solid ground, completely smoothing over the subterranean deposition sites beneath.

CLOTH SIMULATION FILTER (CSF)



The CSF algorithm mathematically strips the rootball vegetation, revealing linear terrain depressions, micro-slumps, and vault roofs invisible to the human eye.

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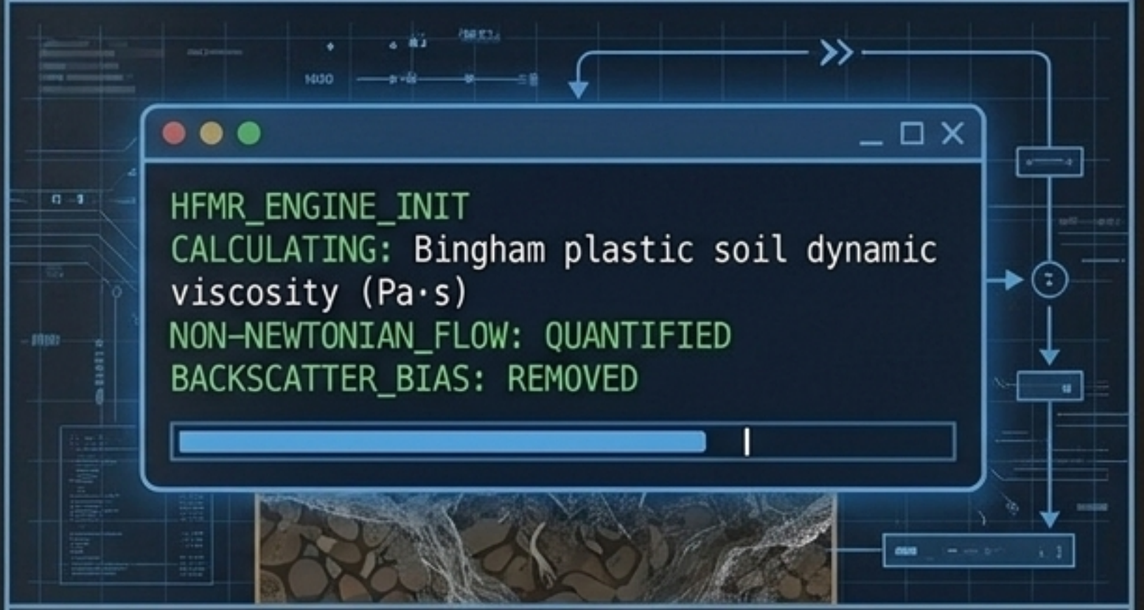
PENETRATING THE SUBSURFACE: BLUE OCEAN SAR

Saturated silt-clay matrices introduce a massive +1.5 dB to +4.0 dB backscatter bias in traditional radar.
We correct the physics to see through the soil.

THE PHYSICS OF SAR



THE RHEOLOGY ENGINE



By quantifying the non-Newtonian flow of the mud, the code automatically removes the dielectric backscatter bias, isolating the precise geometric shape of the concealed artifact.

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THE ENGINE: MATHEMATICAL JOINT-INVERSION (PHYS-UNMASK)

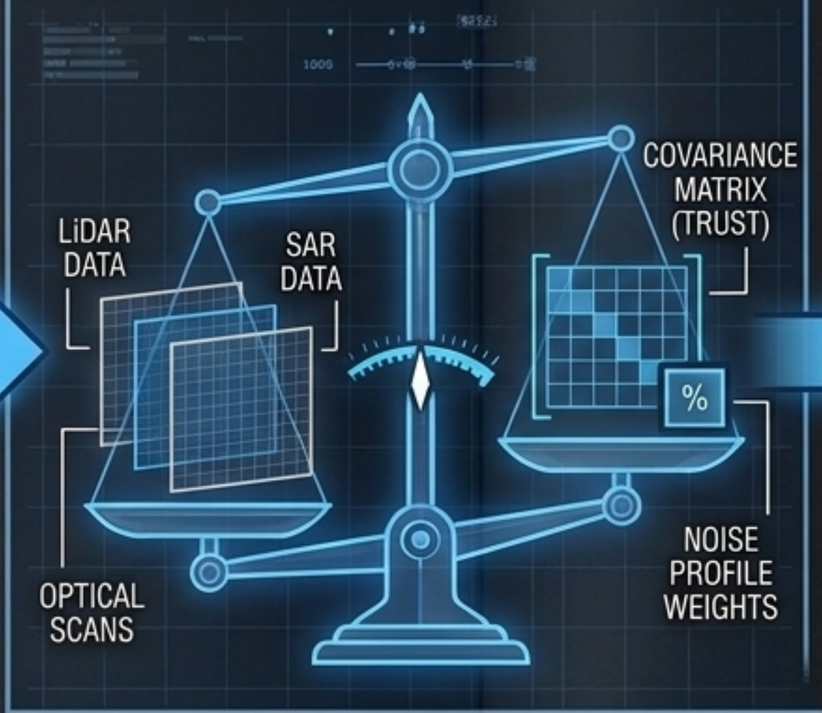
THE LINEAR BASELINE

$$X = AS + N$$



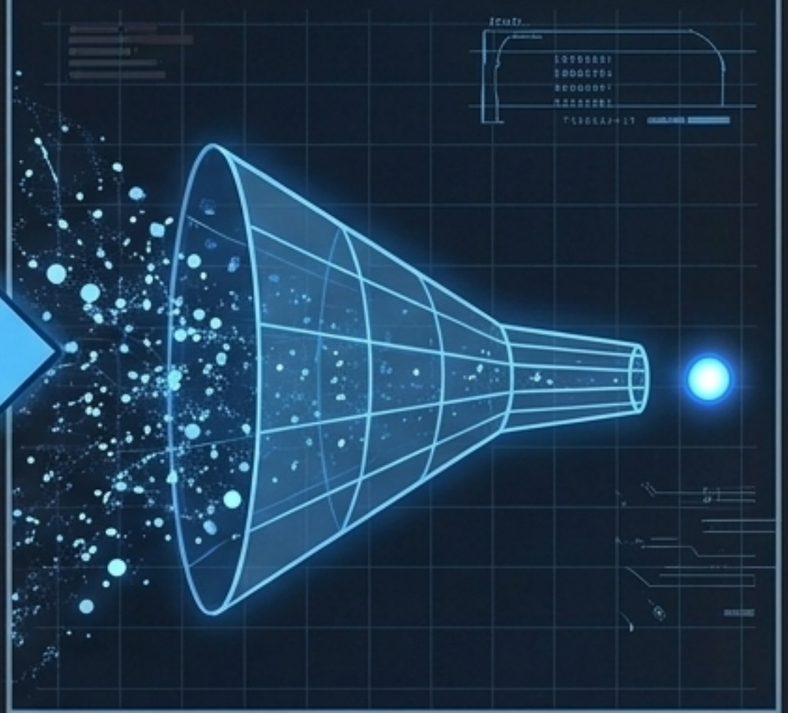
Observed data (**X**) equals the measurement matrix (**A**) times the true environment (**S**) plus unavoidable noise (**N**).

COVARIANCE TRUST WEIGHTING



When fusing LiDAR, SAR, and optical scans, one degraded sensor can corrupt the rest. The Covariance Matrix mathematically balances the impact of each sensor strictly based on its verifiable noise profile, ending subjective human guesswork.

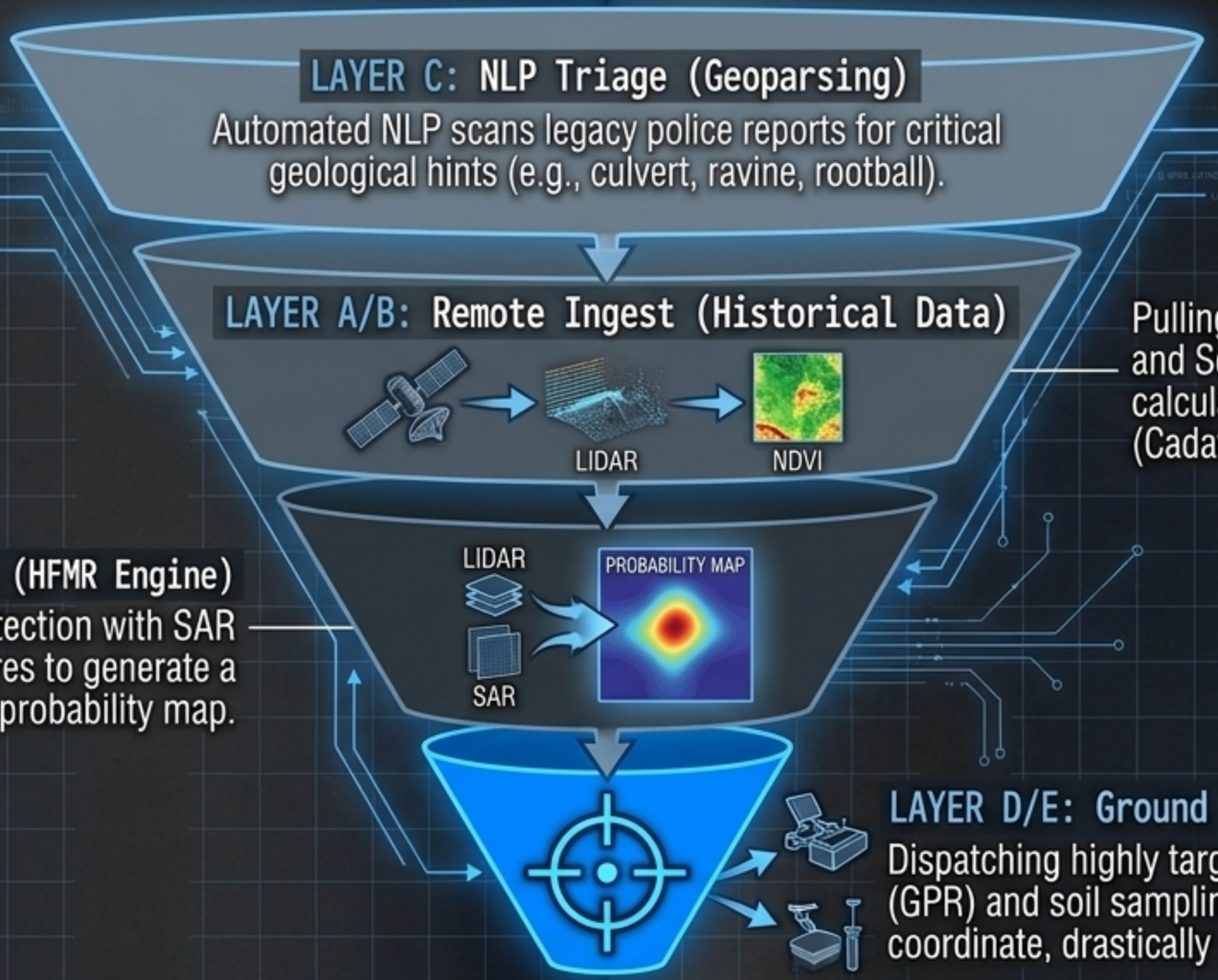
L1-NORM SPARSE REGULARIZATION



By introducing the L1-norm regularization term, the math forces sparsity. It strictly shrinks all background noise variables to exactly zero, isolating the single anomalous forensic target.

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ARCHITECTURE OF THE PIPELINE



"CULVERT FOUND"

"RAVINE_DETECTED"

Pulling existing USGS 3DEP LiDAR and Sentinel-2 optical data to calculate NDVI anomalies (Cadaver Decomposition Islands).

THE CORE: Physics Inversion (HFMR Engine)

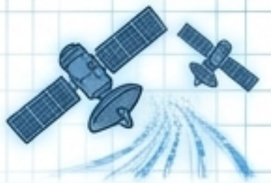
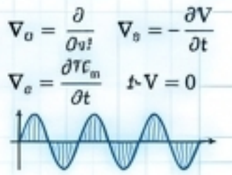
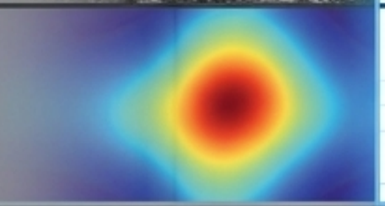
Fusing LiDAR void detection with SAR dielectric signatures to generate a calibrated P(grave|data) probability map.

LAYER D/E: Ground Truth & Confirmation

Dispatching highly targeted Ground-Penetrating Radar (GPR) and soil sampling teams to the exact optimized coordinate, drastically reducing physical search time.

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PARADIGM COMPARISON: HEURISTIC VS. DETERMINISTIC SEARCH

DIMENSION	⚠ LEGACY GSAR (PROBABILISTIC) ⚠	PHYS-UNMASK (DETERMINISTIC PHYSICS)
DATA SOURCE	Requires expensive new drone/K-9 deployments. 	Reprocesses existing historical satellite/3DEP archives. 
PROCESSING METHOD	Visual scanning, heuristic feature extraction, human intuition. 	Evaluates signals against Maxwell's electromagnetic wave equations. 
TREATMENT OF NOISE	Naïve Bayes double-counting inflates false positives. 	Covariance weighting mathematically balances sensor reliability. 
TARGET ISOLATION	Blinded by dense canopy, snow bridging, and rootballs. 	L1-norm sparsity and CSF mathematically strip masking layers. 
OUTPUT	Subjective probability grids. 	Precise, verifiable coordinate bounding boxes. 

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THE BLUE OCEAN REGISTRY: UNTAPPED FRONTIERS

TITLE	BADGES	DESCRIPTION
GRAVESOIL NECROBIOME PTI	Maturity: Emerging; Patent Space: HIGH	Tracks 16S/MPS soil microbiome shifts that persist months after remains are moved.
ADIPOCERE DIELECTRIC FINGERPRINTS	Maturity: Field-Observed; Patent Space: MEDIUM	Library of GPR signatures for grave wax, extending radar usefulness to 15+ year cold cases.
VOC PERSISTENCE CHEMISTRY	Maturity: Published; Patent Space: MED-HIGH	Proving soil VOC profiles discriminate used sites 200+ days after target removal.
HYPERSPECTRAL IMAGING (700-1800 NM)	Maturity: Controlled Studies; Patent Space: LOW-MED	Narrow-band vegetation stress response for clandestine anomaly detection.
LABRADOR-CLASS FIELD E-NOSE	Maturity: Prototype; Patent Space: LOW (Prior Art)	Fusing 12-sensor VOC arrays with probability maps for sub-60 second readings.
MUON TOMOGRAPHY	Maturity: 2025 Demo; Patent Space: LOW	Cosmic-ray attenuation mapping for deep voids/vaults beyond GPR range.

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FINAL VALIDATION: LAYER E (THE BIOLOGICAL PROOF)

The algorithm handles the funnel; physics and biology confirm the find.

THE MECHANISM: qPCR CONFIRMATION ASSAY

Tiling Degraded DNA: Utilizing mtDNA HV1 mini-amplicons (<=160 bp) to test highly degraded soil eDNA.

QUALITY CONTROL GATES

Triplicate concordance ensures exact measurements (Ct spread <= 1.5). Internal Positive Controls (IPC) flag humic acid inhibition.

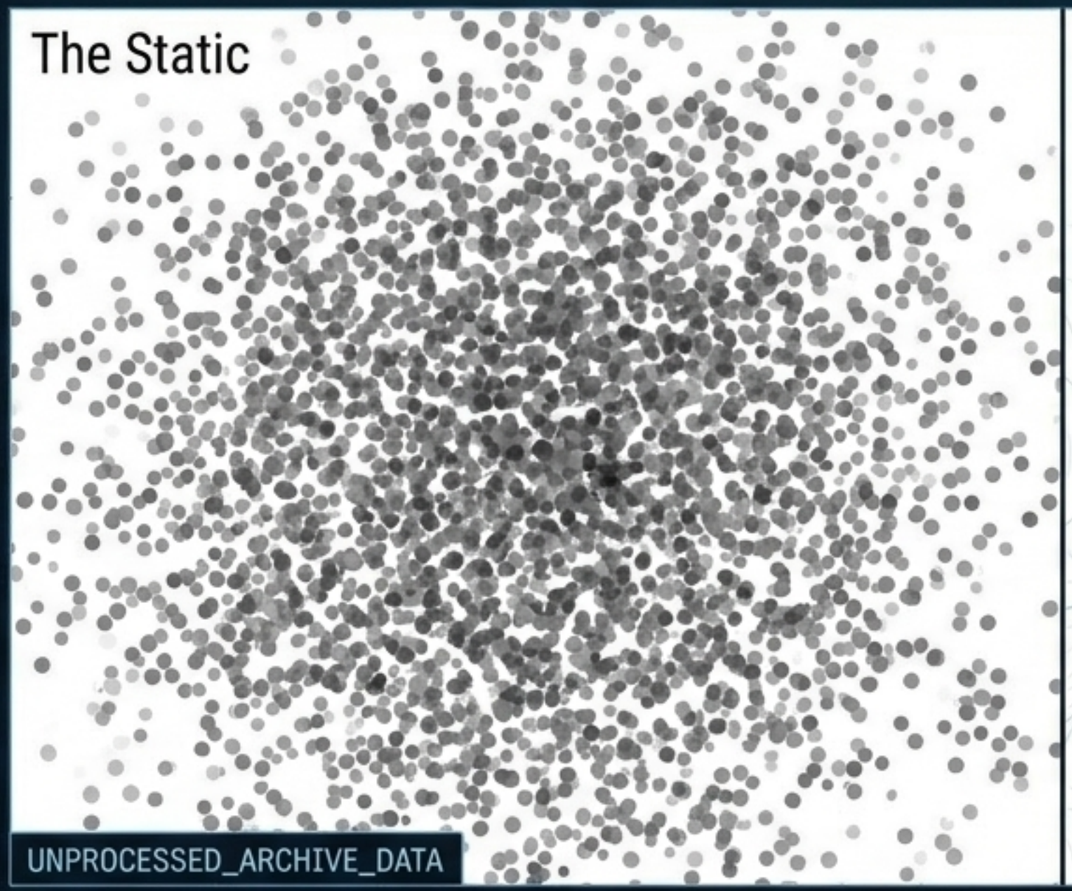
THE OUTPUT: interpret_sample()

A single, defensible call per sample. Reaching the LOD95/LOQ threshold confirms empirical presence of decomposition-derived human DNA, triggering immediate sequencing.

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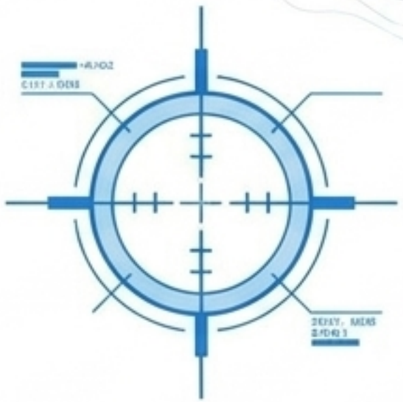
THE ALGORITHM IS THE FIRST RESPONDER

The Static



UNPROCESSED_ARCHIVE_DATA

The Signal



PHYS-UNMASK_DETERMINISTIC_ISOLATION

The future of wilderness recovery isn't fielding more human volunteers into hostile environments.
It is applying relentless, deterministic math to the data we already possess. The code is ready.
We bypass the limitations of human scale. We find the whisper in the static.

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